

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 3

Duration : 1 Hour
Total Marks: 30

Annexure A

Constraint-Based Problem Solving

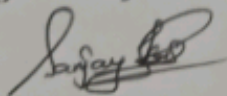
Code of Conduct:

/192521055

I Sarajay Krishnan.P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

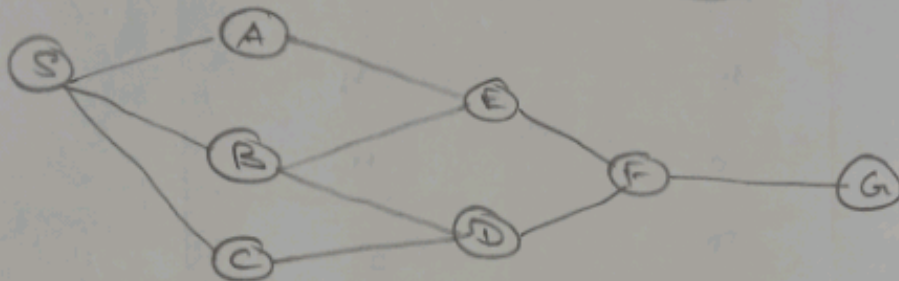
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
- Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Constraint-Based Problem Solving:

1.

Search Algorithm weighted graph.~~Root~~Problem Statement:

consider the following graph. Find the optimal constraints of the given search algorithm through the (D_1, D_2, D_3)

edge	cost
$S \rightarrow A$	2
$S \rightarrow B$	5
$A \rightarrow C$	2
$A \rightarrow D$	4
$B \rightarrow D$	5
$C \rightarrow G$	0
$D \rightarrow G$	3

Heuristic values:

Node	$h(n)$
S	6
A	5
B	4
C	4
D	2
G	0

Formula:

$$f(n) = g(n) + h(n)$$

Solution:

Start at S: $g(S) = 0$

Expand

$$* A: g = 2, f = 2 + 5 = 7$$

$$* B: g = 5, f = 5 + 4 = 9$$

Expand A

$$* C: g = 4, f = 4 + 4 = 8$$

$$* D: g = 6, f = 5 + 4 = 9$$

Expand (C & D)

expand A first (G gets $g = 9$).

Expand D: update G to $g = 6 + 3 = 9$

path cost = 9.

After $S \rightarrow A \rightarrow C \rightarrow G$

$\therefore$ cost = 9

Problem Statement:

It transform the initial state ^{to} goal using
A* search and maintain distance.

$$h(n) = \sum_i (x_i^g - x_i^a | + | y_i^g - y_i^a |)$$

A* evaluates:

$$f(n) = g(n) + h(n)$$

Operation	5L Jug	3L Jug
Start	0	0
Fill 5L Jug	5	0
Pour into 3L Jug	2	3
Empty 3L Jug	2	0
Transfer remain 2L	0	2
Fill 5L Jug	0	2
Pour into 3L Jug until full	- 4	3

3. Uniform Cost Search.

Soln..

1. Start at switch cost 0.
2. Expand S: $A=0$, $B=2$
3. Expand B: $C=3$, $D=9$
4. Expand C: $G=8$

The total cost = 9.